作业：13-23,25,27,30；15-9,11.

第二十六讲（2）

虚功原理与单位载荷法回顾与例题

第十三章　能量法

13.1 外力功与应变能

13.2 功的互等定理

13.3 卡氏定理

13.4 虚功原理与单位载荷法

上讲回顾

变形体虚功原理

满足平衡方程、静力边界条件

满足变形协调条件、位移边界条件

外力 $q(x)$

内力 $M(x)$

静力许可场

虚位移 $w^*(x)$

虚变形 $\mathrm{d}\theta^*$

运动许可场

彼此独立

外力虚功 W_e

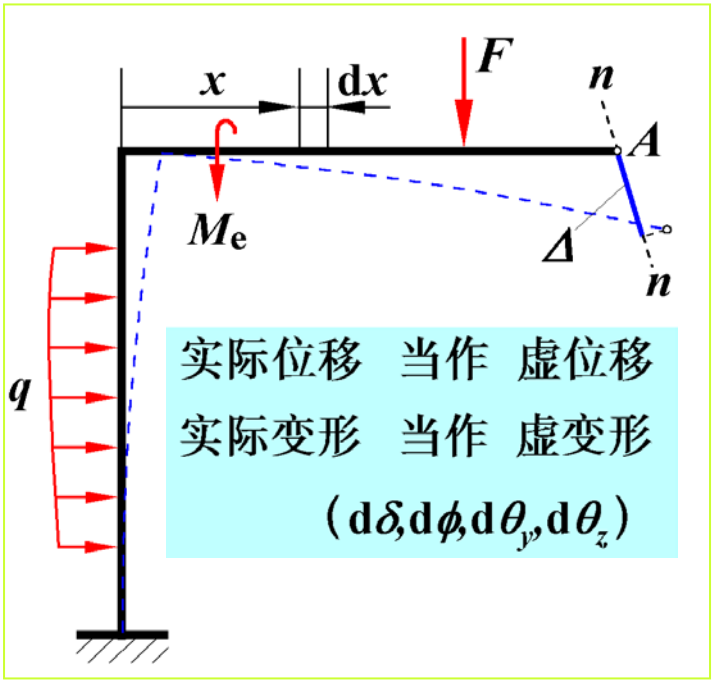
内力虚功 W_i

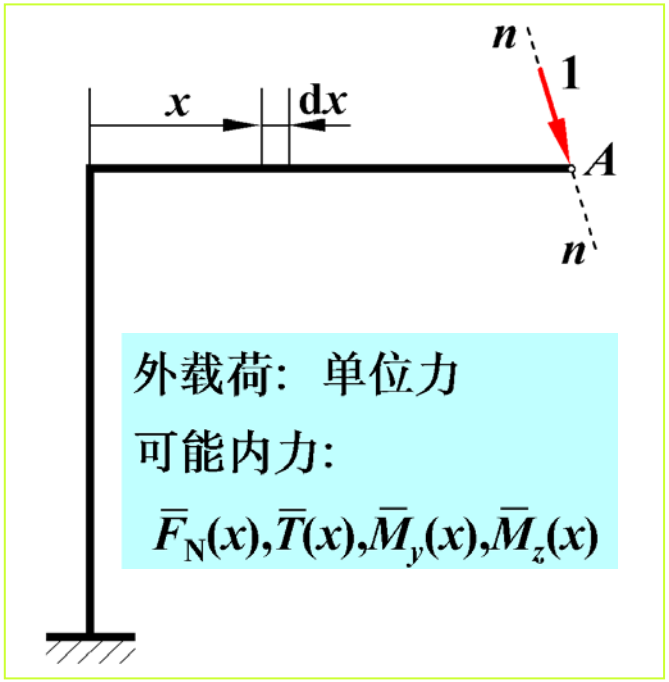
适用于线弹性、非线弹性、非弹性体

上讲回顾

单位载荷法

结构的载荷状态





构造单位（载荷）状态

Δ -- $\mathrm{d}\delta, \mathrm{d}\phi, \mathrm{d}\theta_y, \mathrm{d}\theta_z$ → 运动许可场

1 -- $\bar{F}_N, \bar{T}, \bar{M}_y, \bar{M}_z$ → 静力许可场

$$\Delta = \int_l \bar{F}_N(x)\mathrm{d}\delta + \int_l \bar{T}(x)\mathrm{d}\phi + \int_l \bar{M}_y(x)\mathrm{d}\theta_y + \int_l \bar{M}_z(x)\mathrm{d}\theta_z$$

上讲回顾

单位载荷法

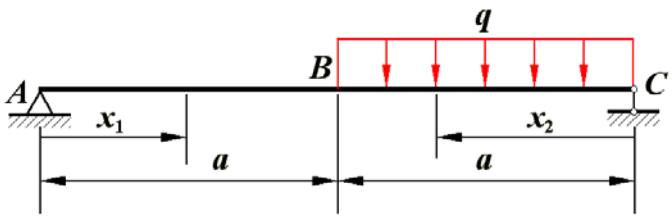
线弹性桁架

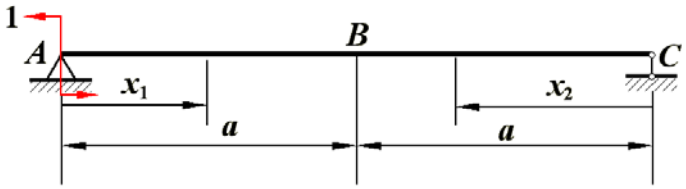
$$\Delta=\sum_{i=1}^{n}\frac{\overline{F}_{\mathrm{N}i}F_{\mathrm{N}i}l_i}{E_iA_i}$$

线弹性组合变形梁和刚架

$$\Delta=\int_l\frac{\overline{T}(x)T(x)}{GI_{\mathrm{t}}}\mathrm{d}x+\int_l\frac{\overline{M}_y(x)M_y(x)}{EI_y}\mathrm{d}x+\int_l\frac{\overline{M}_z(x)M_z(x)}{EI_z}\mathrm{d}x$$

- 正确施加单位载荷，求相应位移；
- 画出载荷状态、单位状态受力图；
- 分段、坐标、内力符号保持一致；
- 桁架问题——列表；
- 图形互乘法——弯矩图互乘求积分。





$$\int_l\overline{M}(x)M(x)\mathrm{d}x$$

上讲回顾

卡氏定理与单位载荷法

线弹性结构

$$\Delta_k = \frac{\partial V_\varepsilon}{\partial F_k} = \sum_{i=1}^{n} \frac{F_{Ni} l_i}{E_i A_i} \frac{\partial F_{Ni}}{\partial F_k} \qquad \Delta = \sum_{i=1}^{n} \frac{\overline{F}_{Ni} F_{Ni} l_i}{E_i A_i}$$

$$\Delta_k = \frac{\partial V_\varepsilon}{\partial F_k} = \int_l \frac{T(x)}{GI_t} \frac{\partial T(x)}{\partial F_k} \mathrm{d}x + \int_l \frac{M(x)}{EI} \frac{\partial M(x)}{\partial F_k} \mathrm{d}x$$

$$\Delta = \int_l \frac{\overline{T}(x) T(x)}{GI_t} \mathrm{d}x + \int_l \frac{\overline{M}(x) M(x)}{EI} \mathrm{d}x$$

非线弹性结构

$$\Delta_k = \frac{\partial V_C}{\partial F_k}$$

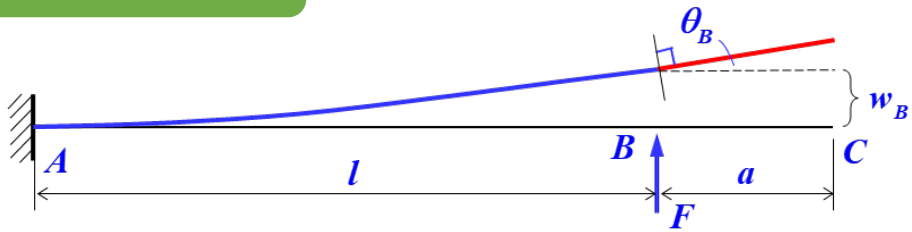
克罗蒂-恩格塞定理

$$\Delta = \int_l \overline{F}_{\mathrm{N}}(x) \mathrm{d}\delta + \int_l \overline{T}(x) \mathrm{d}\phi + \int_l \overline{M}(x) \mathrm{d}\theta$$

上讲回顾

影响系数与单位载荷法

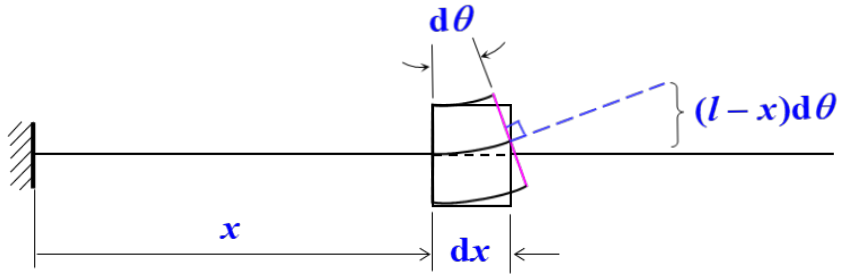
$$\mathrm{d}\theta_B = \mathrm{d}\theta, \quad \mathrm{d}w_B = (l-x)\mathrm{d}\theta$$



影响系数——微段变形的贡献

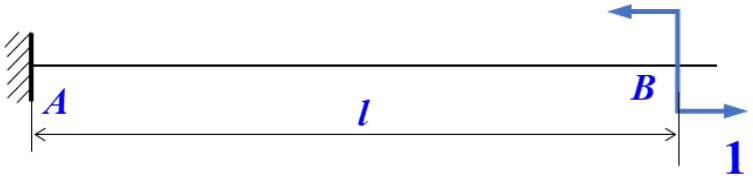
$$\mu_{\theta B} = 1, \quad \mu_{wB} = l - x$$

$$\theta_B = \int_0^l \mu_{\theta B}\,\mathrm{d}\theta, \quad w_B = \int_0^l \mu_{wB}\mathrm{d}\theta$$

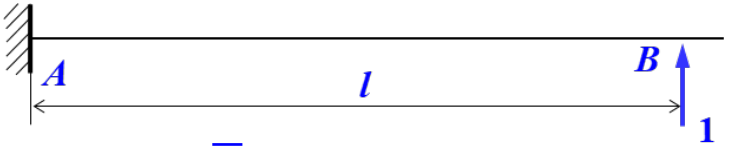


微段变形累加形成截面挠度与转角

- 影响系数就是单位载荷引起的可能内力；
- 由几何关系可以导出单位载荷法，与材料性质无关；
- 能量法无需具体的几何分析，间接、抽象，解决复杂问题。



$$\bar{M}_1(x) = 1 = \mu_{\theta B}$$



$$\bar{M}_2(x) = l - x = \mu_{wB}$$

$$\frac{\partial M_2(x)}{\partial F} = l - x = \mu_{wB}$$

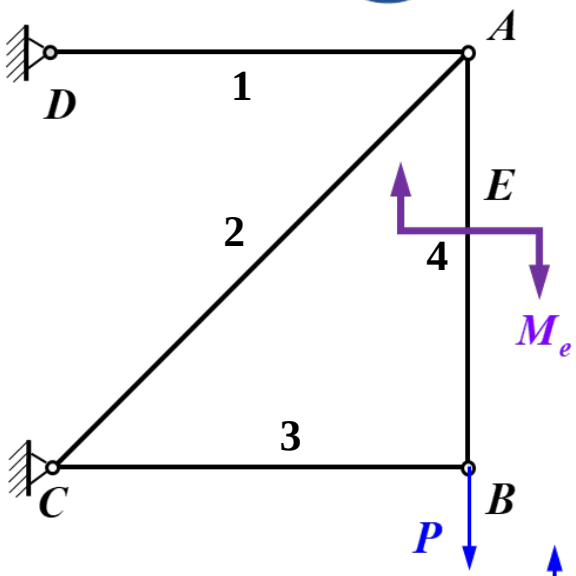
也是影响系数

思考题

任意处E加力矩M_e

1） θ_{AB}是M_e的相应位移吗？

2） $\theta_{AB}=\dfrac{\partial V_\varepsilon}{\partial M_e}$ ？

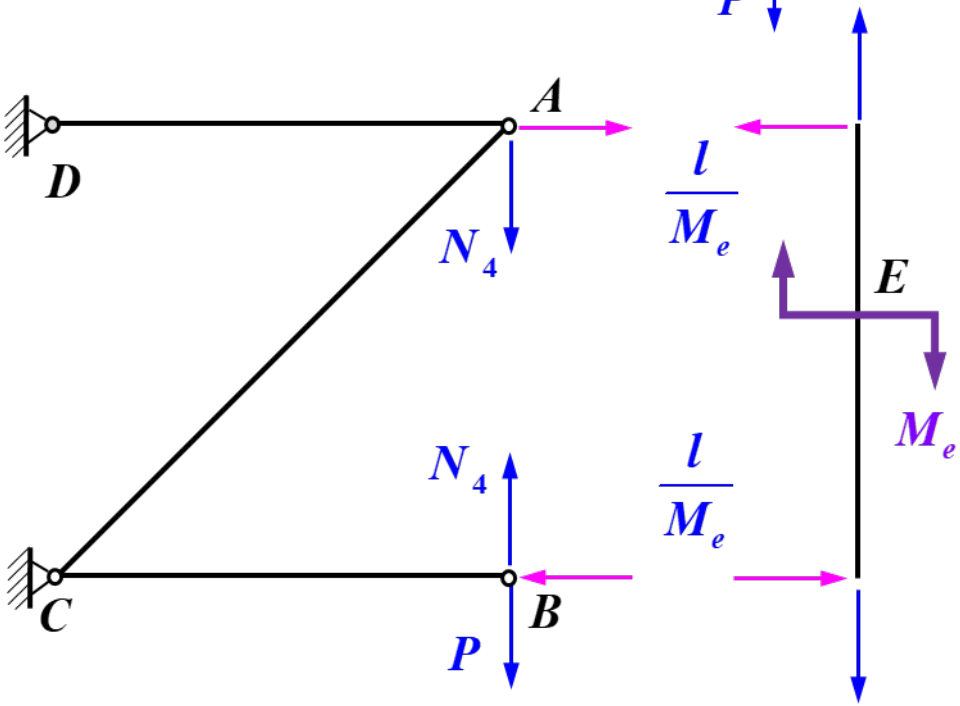


$$\frac{\partial V_\varepsilon}{\partial M_e}=\theta_E$$

当$M_e=Fl$， 各杆轴力不变，

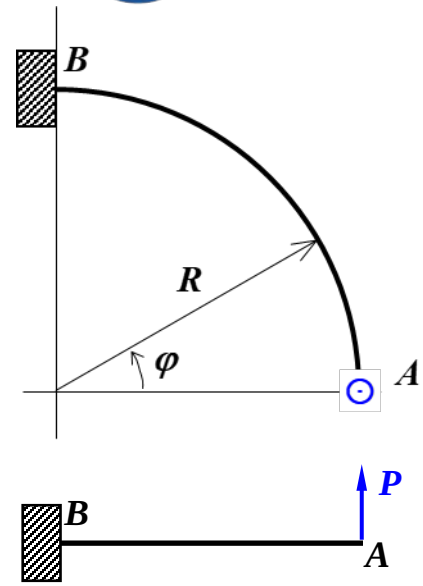
$\theta_E=\theta_{AB}$

AB杆无弯曲，$\theta_E=\theta_{AB}$



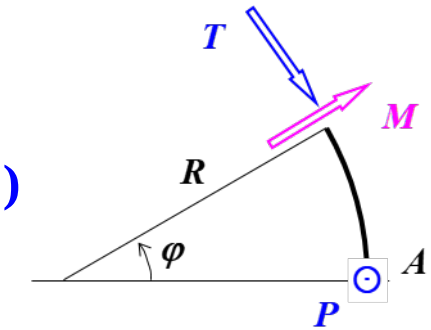
单位载荷法的应用

例13-21 图示1/4圆弧小曲率杆AB，EI和GJ_k为常数，受载荷P。求：挠度f_A，扭转角ϕ_A



解：

载荷状态<P>



$$M(\varphi)=PR\sin\varphi,\quad T(\varphi)=PR(1-\cos\varphi)$$

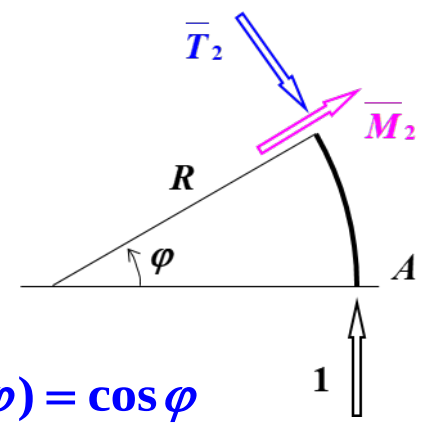
单位载荷状态<1>

$$\overline{M}_1(\varphi)=R\sin\varphi,\quad \overline{T}_1(\varphi)=R(1-\cos\varphi)$$

$$f_A=\int_0^{\frac{\pi}{2}}\frac{\overline{M}_1(\varphi)M(\varphi)}{EI}R\,\mathrm{d}\varphi+\int_0^{\frac{\pi}{2}}\frac{\overline{T}_1(\varphi)T(\varphi)}{GJ_k}R\,\mathrm{d}\varphi$$

$$=\frac{\pi PR^3}{4EI}+\left(\frac{3\pi}{4}-2\right)\frac{PR^3}{GJ_k}(\uparrow)$$

单位载荷状态<2>

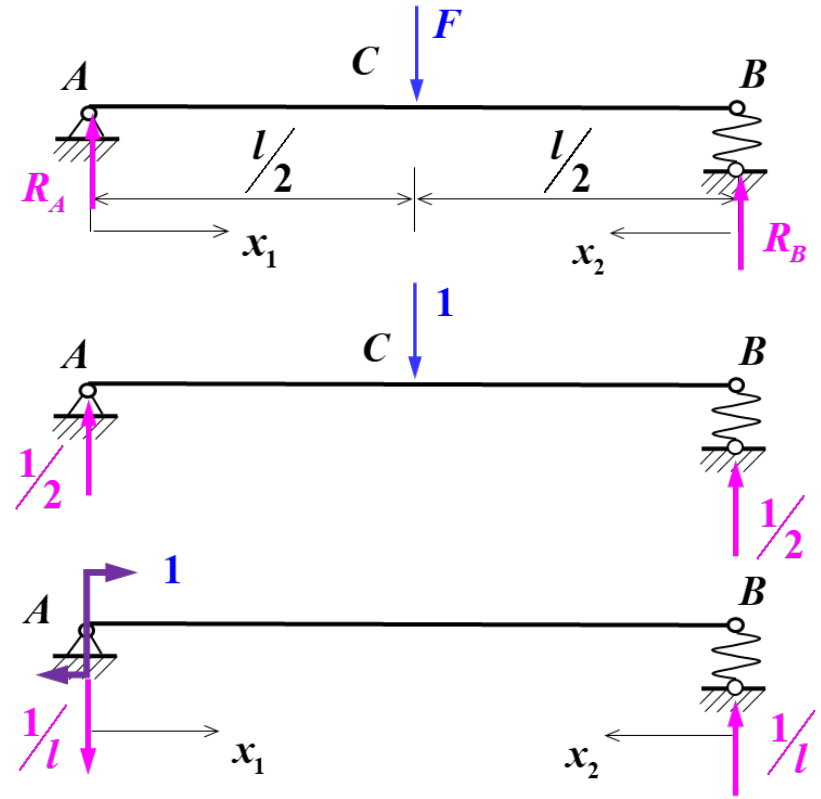


$$\overline{M}_2(\varphi)=-\sin\varphi,\quad \overline{T}_2(\varphi)=\cos\varphi$$

$$\theta_A=\frac{PR^2}{4EI}+\left(1-\frac{\pi}{4}\right)\frac{PR^2}{GJ_k}$$

单位载荷法的应用

例13-22 图示一端弹簧支撑的简支梁AB，EI为常数，弹簧刚度系数k。求：f_C，θ_A



解：载荷状态 $R_A = R_B = \dfrac{F}{2}$

$$M_1(x_1) = \frac{F}{2}x_1, \quad M_2(x_2) = \frac{F}{2}x_2, \quad N = \frac{F}{2}$$

单位载荷状态

$$\frac{\partial M_1}{\partial F}, \quad \frac{\partial N}{\partial F}$$

$$\overline{M}_{11}(x_1) = \frac{x_1}{2}, \quad \overline{N}_1 = \frac{1}{2}$$

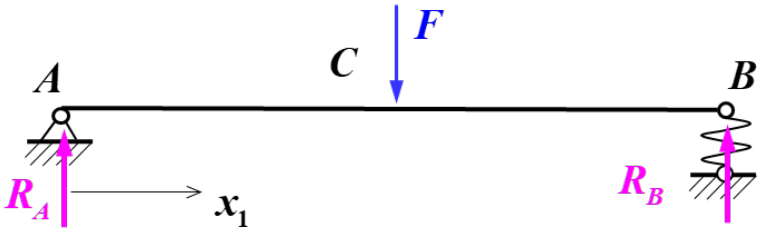
$$f_C = 2\int_0^{\frac{l}{2}} \frac{\overline{M}_{11}M_1}{EI}\mathrm{d}x_1 + \overline{N}\frac{N}{k} = \frac{Fl^3}{48EI} + \frac{F}{4k}(\downarrow)$$

$$\overline{M}_{12}(x_1) = 1 - \frac{x_1}{l}, \quad \overline{M}_{22}(x_1) = \frac{x_2}{l}, \quad \overline{N}_2 = \frac{1}{l}$$

$$\theta_A = \int_0^{\frac{l}{2}} \frac{\overline{M}_{12}M}{EI}\mathrm{d}x_1 + \int_0^{\frac{l}{2}} \frac{\overline{M}_{22}M}{EI}\mathrm{d}x_2 + \overline{N}_2\frac{N}{k} = \frac{Fl^2}{16EI} + \frac{F}{2lk} \quad (\circlearrowright)$$

单位载荷法的应用

例13-22 图示一端弹簧支撑的简支梁AB，EI为常数，弹簧刚度系数k。求：f_C，θ_A

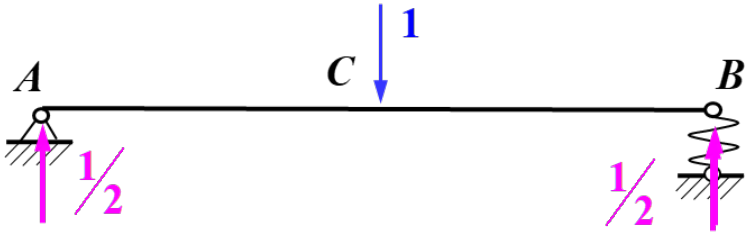


解：若以梁为对象

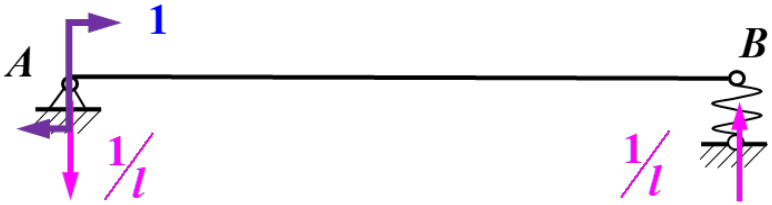
$$R_A = R_B = \frac{F}{2}$$

$$M_1(x_1) = \frac{F}{2}x_1, \quad w_B = \frac{F}{2k}(\downarrow)$$

$$\overline{M}_{11}(x_1) = \frac{x_1}{2}$$



$$1\times f_C - \frac{1}{2}w_B = 2\int_0^{\frac{l}{2}} \frac{\overline{M}_{11}M_1}{EI}\mathrm{d}x_1$$



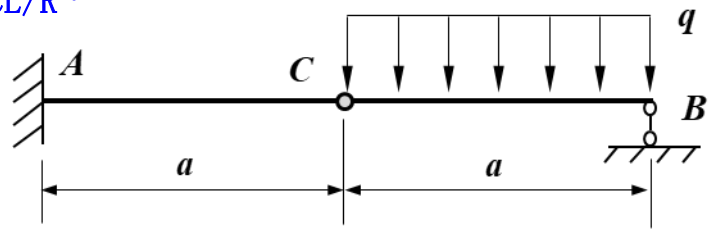
$$\overline{M}_{12}(x_1) = 1 - \frac{x_1}{l}, \quad \overline{M}_{22}(x_2) = \frac{x_2}{l}$$

$$1\times\theta_A - \frac{1}{l}w_B = \int_0^{\frac{l}{2}} \frac{\overline{M}_{12}M_1}{EI}\mathrm{d}x_1 + \int_0^{\frac{l}{2}} \frac{\overline{M}_{22}M_2}{EI}\mathrm{d}x_2$$

外力虚功等于内力虚功

单位载荷法的应用

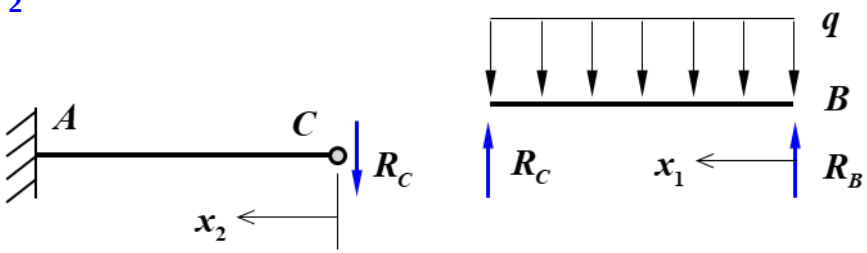
例13-23 图示组合梁，EI为常数，求：f_C, $\theta_{CL/R}$.



解： 载荷状态<P>

$$R_B = R_C = \frac{qa}{2}$$

$$M_1(x_1) = R_B x_1 - q x_1 \frac{x_1}{2}, \quad M_2(x_2) = -R_C x_2$$

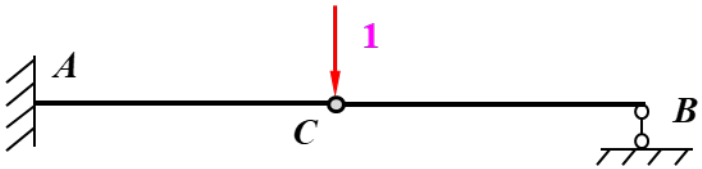


单位载荷状态<1>

$$\overline{M}_{11} = 0,$$

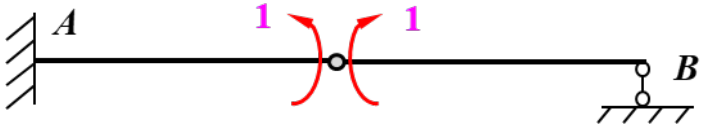
$$\overline{M}_{21} = -x_2$$

$$f_C = \int_0^a \frac{\overline{M}_{21} M_2}{EI} \mathrm{d}x_2 = \frac{R_C l^3}{3EI} = \frac{qa^4}{6EI} (\downarrow)$$



单位载荷状态<2>

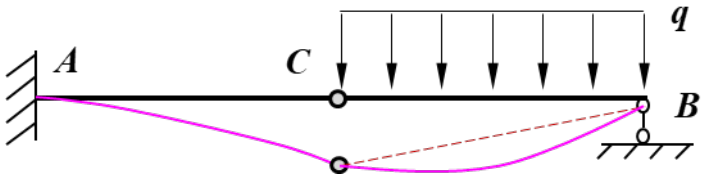
$$\overline{M}_{12} = \frac{x_1}{a}, \quad \overline{M}_{22} = 1 + \frac{x_2}{a}$$



$$\theta_{CL/R} = \int_0^a \frac{\overline{M}_{12} M_1}{EI} \mathrm{d}x_1 + \int_0^a \frac{\overline{M}_{22} M_2}{EI} \mathrm{d}x_2$$

$$= \frac{qa^3}{24EI} - \left(\frac{qa^3}{4EI} + \frac{qa^3}{6EI}\right) = -\frac{3qa^3}{8EI} \quad (\circlearrowright\circlearrowleft)$$

右顺　左顺　右逆



本讲：虚功原理与单位载荷法回顾

外力虚功等于内力虚功

下讲：静不定问题分析